

DAMON KNIGHT'S Best Short Novel — DIO

infinity PDC

SCIENCE FICTION

Sept

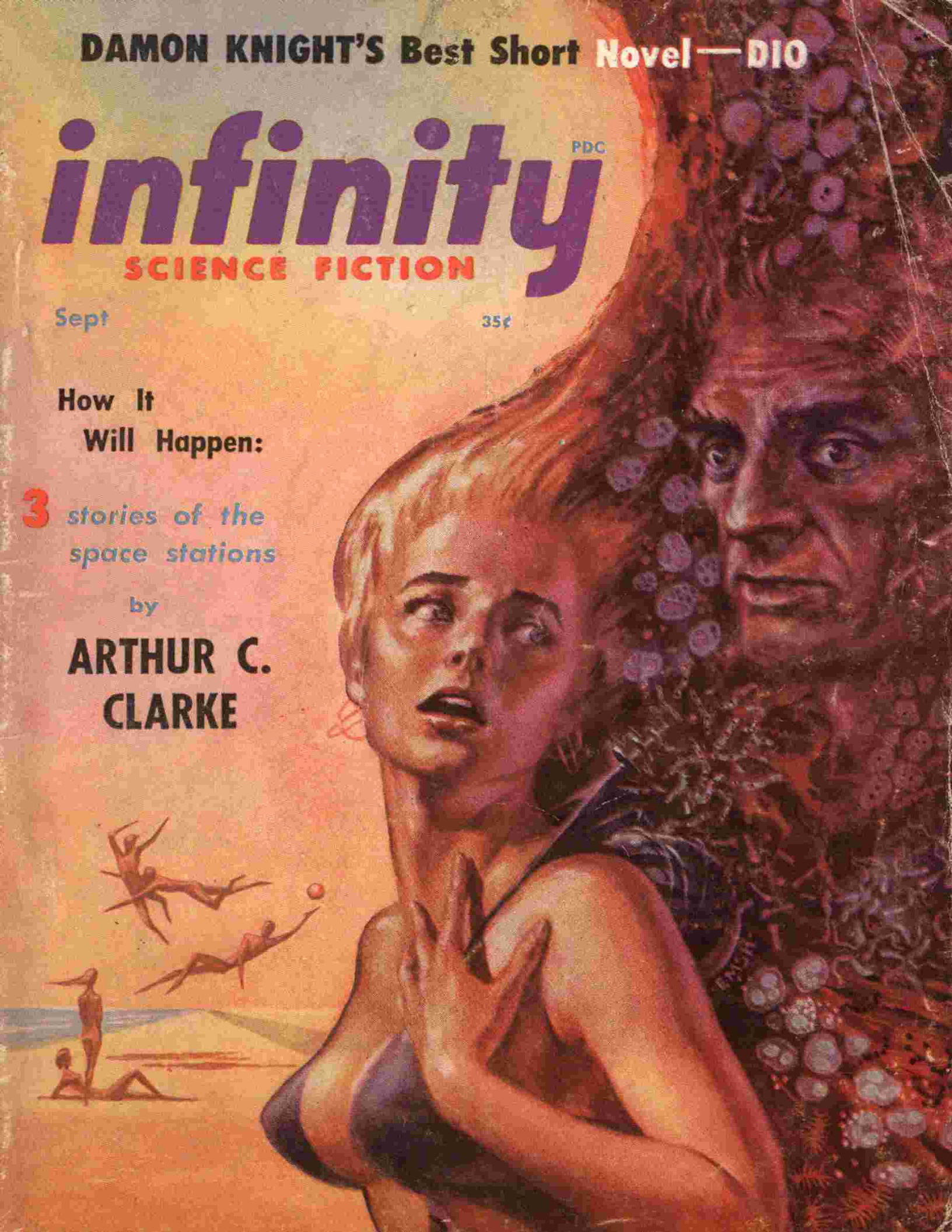
35¢

**How It
Will Happen:**

3 *stories of the
space stations*

by

**ARTHUR C.
CLARKE**



DAMON KNIGHT'S Best Short Novel — DIO

infinity PDC

SCIENCE FICTION

Sept

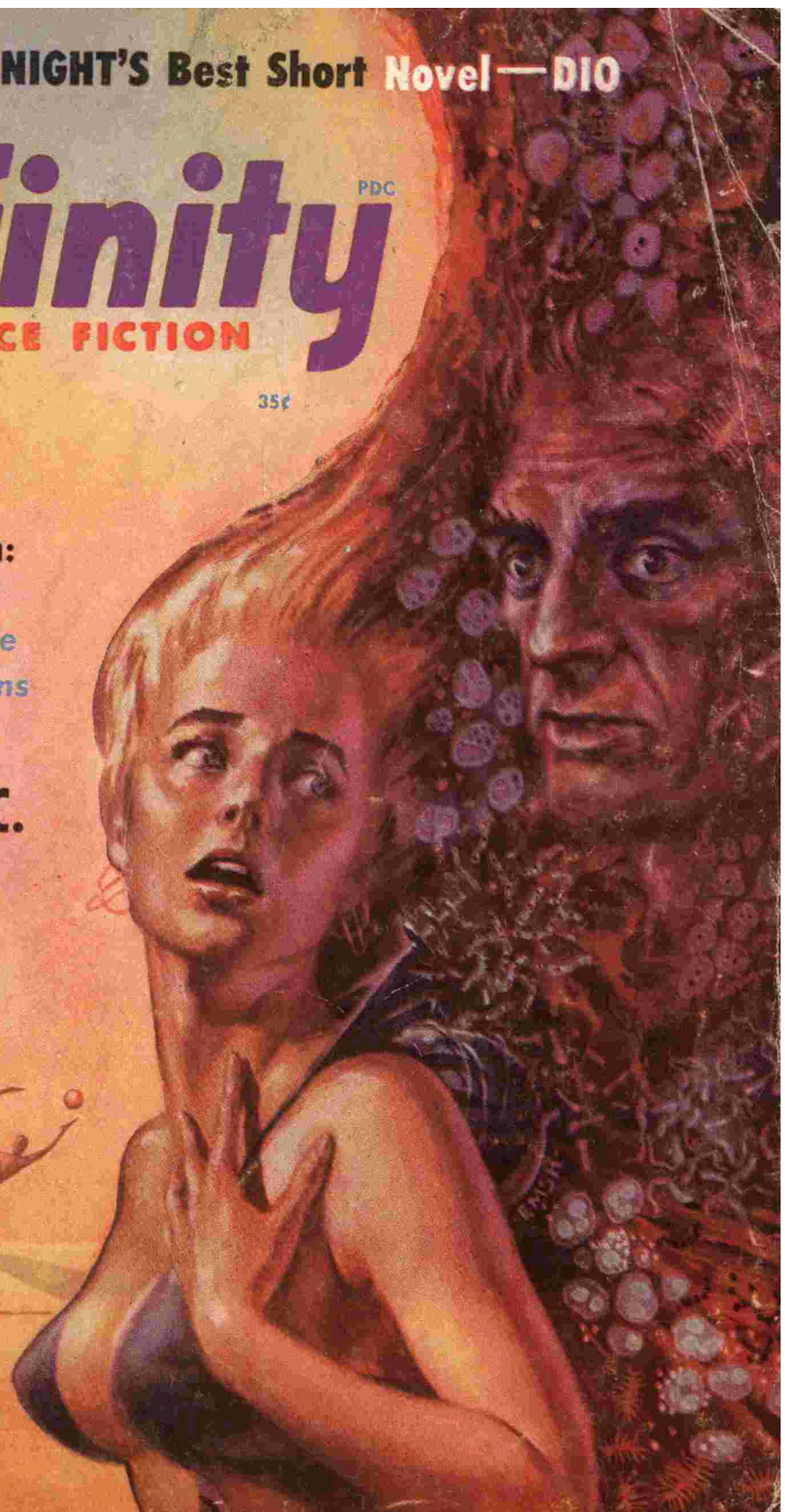
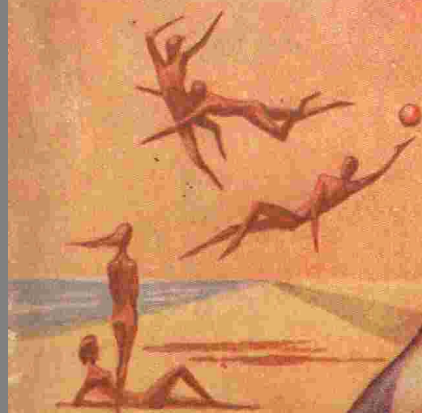
35¢

**How It
Will Happen:**

3 stories of the
space stations

by

**ARTHUR C.
CLARKE**



The Project Gutenberg eBook of Earth transit

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Earth transit

Author: Charles L. Fontenay

Illustrator: Richard Kluga

Release date: September 7, 2023 [eBook #71589]

Language: English

Original publication: New York, NY: Royal Publications, Inc., 1957

Other information and formats: www.gutenberg.org/ebooks/71589

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK EARTH
TRANSIT ***

Earth Transit

By CHARLES L. FONTENAY

Illustrated by KLUGA

**When murder occurs on a spaceship,
the number of suspects is at an absolute
minimum—and Lefler was that minimum!**

**[Transcriber's Note: This etext was produced from
Infinity September 1957.**

**Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

The centerdeck chronometer said 1840 hours.

That startled Lefler into full wakefulness. He was forty minutes overdue in relieving Makki in the control room.

That wasn't like Makki, he thought as he pulled on his coveralls hastily. Makki was as punctual—and as thorough—as the maze of machinery whose destiny he guided. He was as cold as that machinery, too, when others made a mistake. It made him an efficient spaceship captain and a disliked man.

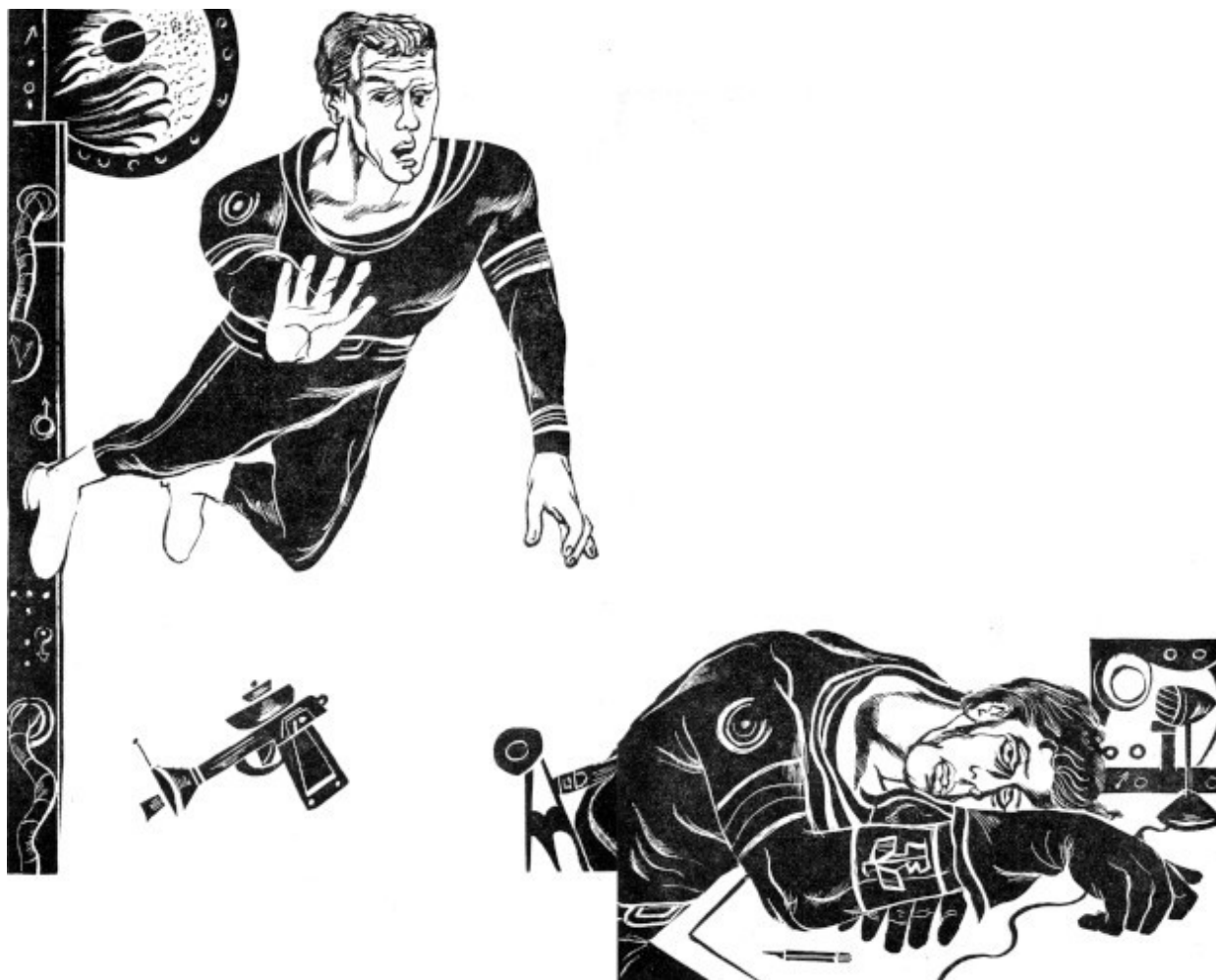
Lefler shook his head to clear it of dream-haunted memories. He had awakened from a nightmare in which, somewhere, there was angry shouting, to find himself floating midway from floor to ceiling of the centerdeck of the *Marsward IV*. Somehow, his retaining straps had become unbuckled, letting him float free of his bunk in his sleep.

Not pausing to fold his bunk back against the curving hull, Lefler made his way briskly up the companionway, through the empty and darkened astrogration deck and into the control room.

"Makki," he called to the figure reclining in the control chair. "Makki, I'm due to relieve you. You're forty minutes overtime."

There was no answer. Floating up to the control chair, Lefler recoiled, bouncing painfully off the automatic pilot.

Makki was dead. Death had robbed his wide eyes of their dark scorn and smoothed the bitter lines of his heavy face. His coveralls were charred around the heat-beam burn in his chest.



The heat-gun bumped against Lefler's shoulder and drifted away at an angle across the gravityless control room. Lefler stared after it in horror.

Licking dry lips, he punched the communicator button.

"Blue alert!" he croaked into the microphone. "All hands to control room. Blue alert!"

Anchoring himself to the automatic pilot, he studied Makki's body as dispassionately as he could. The captain was still strapped in the cushioned chair. Oddly, he was wearing gloves.

The log-tape was in the recorder beside the control chair. Clipped to a metal leaf on the stanchion beside the chair was Makki's notepad. Scrawled on it in

the captain's handwriting was the notation: "73rd day. Earth transit."

"What's up, Lefler?" asked a voice behind him. Lefler turned to face Taat, the ship's doctor. Taat, a plump, graying man, was wiping his hands on the white smock he wore.

Lefler moved aside, letting Taat see Makki's body. Taat's eyes widened momentarily, then narrowed with a professional gleam. He stepped quickly to Makki's side, made as if to pick up the dead captain's wrist, then turned back to Lefler with a fatalistic flick of his hands.

"What was it, Lefler?" he asked in a low voice. "A fight?"

"I don't know," said Lefler. "I found him that way."

Taat raised his eyebrows.

"Robwood?" he asked softly.

Robwood's head poked up through the companionway, and he floated into the control room. There was a streak of grease across the engineer's thin face.

"Great space!" exclaimed Robwood at once. "What happened to Makki?"

"Obviously, he's been shot," said Lefler in an even voice. "Any idea who did it, Robwood?"

"Wait a minute," objected Taat mildly. "That sounds like you are accusing Robwood, Lefler."

"I'm not," said Lefler hastily. "I'm not leaving you out, Taat. But there are only the three of us. One of you must have killed him."

"Great space, you don't think that I—" began Robwood.

"Just to get the record straight, Lefler," interrupted Taat, "let's put it this way: one of the three of us must have killed him."

It was not only Lefler's duty watch; as astrogator, he became acting captain as a result of Makki's death. Moving to the side of the dead Makki, he turned the ship's radio transmitter toward distant Earth and pressed the sending key.

"*Marsward IV* to White Sands," he called. "*Marsward IV* to White Sands."

It would be several minutes before a reply could reach them.

Taat, on the other side of the control chair, was examining Makki's corpse. Robwood stood peering over his shoulder.

Lefler waited to see which one would comment first on the fact that Makki was wearing gloves. Neither appeared to notice it.

But the gloves put a thought into Lefler's own mind. Fingerprints!

He looked around the control room and found the heat-gun, bumping against the celestial camera. He pushed himself across the room, pulling a handkerchief from the back pocket of his coveralls as he did so. He wrapped the heat-gun in the handkerchief, stuck it in a drawer beneath one of the control panels, locked the drawer and put the key in his pocket.

The loudspeaker buzzed.

"*Marsward IV*, this is Capetown," said a slightly wavery voice. "We're relaying you to White Sands. Go ahead, please."

Lefler picked up the mike.

"*Marsward IV* to White Sands," he said. "This is Lefler, astrogator. Makki, captain, shot to death under unknown circumstances. I am assuming command. Instructions, please."

Taat turned away from Makki's body.

"He's been dead about thirty minutes." Taat looked at the control room chronometer. It said 1906 hours. "I'm going to list the time in the death certificate as 1830."

"You can tell?" asked Robwood in astonishment.

"By the eyes," said Taat.

"Wait a minute," said Lefler. "It was only 1840 when I started up here. You mean he'd been dead only ten minutes then? He was already forty minutes overdue waking me for my duty watch."

"Could be ten or fifteen minutes either way," conceded Taat. "If he was late, don't forget that we don't know what happened up here."

"One of us does," reminded Lefler grimly.

"Capetown to *Marsward IV*," said the loudspeaker. "Relaying instructions from White Sands. Lefler's temporary command of ship confirmed. All personnel will be booked on suspicion of murder and mutiny on arrival at Marsport. Captain Makki's body will be preserved and brought down at Marsport. Each crew member will dictate a statement on the circumstances of Captain Makki's death and an outline of his past association with Captain Makki, separately, on this beam for relay to Marsport."

The three looked at each other.

"That's that," said Lefler. "Robwood, if you and Taat will take Makki's body away and secure it outside the airlock, I'll get the ship's records up to date."

Taat unbuckled Makki's body from the control chair. It did not change its slightly bent position as it drifted slowly upward.

"Why do you reckon he's wearing gloves, Lefler?" Taat asked curiously.

"I wondered when one of you fellows was going to say something about that!" burst out Robwood, a curious break in his voice. "All of us have been glaring at each other, suspecting each other, when Makki could have committed suicide!"

"Makki?" retorted Lefler dryly. "I doubt it."

Pushing Makki's body down the hatch toward the airlock at the other side of the personnel sphere would have been an easy task for one man, but Lefler wanted Taat and Robwood to watch each other. He didn't want an "accidental" push to send the prime bit of evidence drifting away into space. When they had disappeared down the hatch with the corpse, he eased himself into the control chair and played back the log from the end of Robwood's last shift at 1000 hours.

Makki had recorded the usual observations of the solar, stellar and planetary positions when he went on duty. There was nothing else on the tape.

Lefler stared gloomily at the silent log-recorder. It seemed incredible to him that never again, except on tape, would he hear Makki's harsh, sardonic

voice. The almost inaudible hum of machinery deep in the ship only emphasized the oppressive stillness of space outside its thin walls.

With a sigh, he picked up the log-recorder microphone and pulled the star sextant down to eye level. He would record the bare facts of Makki's death after the initial position observations.

"*Marsward IV*, bound Marsport from White Sands," he recited in a monotone. "Earth-time, October 29, 2048, 1931 hours. Lefler reporting for duty and assuming command as per conversation with White Sands, to be recorded this date."

He squinted into the sextant.

"Positions: Sun-Mars, 24°28'42". Sun-Earth—"

He broke off. Where was Earth? Then he remembered.

"Damn!" he muttered. "The transit! A murder sure messes up the records around here."

The Earth transit was an event of considerable importance to an astrogator on a hop between Earth and Mars. Marsbound it began on the 73rd day out, Earthbound on the 187th day. Timing it, spaceship observers not only checked the accuracy of the ship's orbit, but also contributed data to the mass of knowledge available on the movements of Earth and Mars.

Lefler found the black disc of Earth in the smoked glass that automatically fell across the sextant lens when it swept by the sun. He checked the angle between the black spot and the leading edge of the solar disc.

"Earth transit already under way," he said into the mike. "Angle with leading edge, two minutes, forty seconds...."

He went around the sky, recording planetary and key stellar positions. He had just finished and switched the tape of his conversation with Earth to record in the log when Taat and Robwood returned.

"Makki's body will keep out there as well as in a refrigerator," said Taat with evident satisfaction. "Robwood tied the airlock into the alarm system so nobody can go out and cut the body free without arousing the others."

"You're both mighty cooperative for one of you to be a murderer," remarked Lefler.

"Maybe neither of us is," said Robwood. "As far as I'm concerned, you may be the man."

"Or, as Robwood suggested earlier, Makki may have shot himself," added Taat.

"Robwood, you and I are going to have to do twelve-hour watches from here to Mars, since Taat doesn't know how to operate the controls," said Lefler. "I'll stay on duty till 0600, and you'd better get some sleep after you've radioed your statement to White Sands."

"Okay," said Robwood. "But are we still going to record star positions in the log every eight hours, or just every twelve hours now?"

"Twelve, I think. But the Earth transit's on right now, and until Terra swings across that half a degree of the sun's face, we'd better take readings on that every four hours, anyhow."

"Well, that's just for a little more than two days," said Robwood. "Look, Lefler, I'm overdue on my sleeping time anyway, so how about letting me make my statement on ... on Makki first?"

"Blast away," said Lefler. "The mike's yours. We'll leave the control room so you'll feel freer to talk."

Lefler munched thoughtfully on a hot sandwich. Across the control room, in the astrogator's chair, Taat sucked at a bulb of coffee.

"Nice of you to fix up this lunch, Taat," said Lefler. "I'm not tied strictly to the control room during my watch, you know. But little things like this relax the tension."

"Yes, it's a peculiar situation, Lefler," said Taat in a tone that indicated he had been thinking about it. "Psychologically, I mean. Now if there were only the two of us, and Makki drifting out there dead, both of us would know who shot him. With three of us, it's different.

"You and I are sitting here talking as though neither of us killed Makki. Maybe you hadn't thought of it, but that means that tacitly, for now, we're

assuming Robwood killed him. But, for all I know, you did. And, if you didn't, for all you know, I did."

"Until we find out, I have to suspect you both," said Lefler flatly.

"I could say the same thing," murmured Taat. "But one of us may be lying."

"Of course, Makki could have shot himself, as Robwood suggested," said Lefler. "If he had relaxed his grip on the heat-gun after pressing the trigger, it would have drifted up away from him. There were the gloves, you know."

"Why wouldn't Makki want his fingerprints on the gun if he were committing suicide?" objected Taat. "I'll concede that Makki had strong sadistic tendencies, but my guess is that the murderer put those gloves on him just to raise the possibility of suicide."

Taat finished his coffee and left the control room. Lefler washed down the last bit of his sandwich with his own coffee and called White Sands on the radio. When he received an acknowledgment after the inevitable delay, he began to dictate his statement.

Lefler told of waking from his sleep period and finding himself forty minutes late for his watch. He described his discovery of Makki's body, what followed, and everything he could remember of what Taat and Robwood had said when they came to the control room.

"Makki was thoroughly detested by every member of the crew," Lefler related. "He did not fraternize and no one wanted to fraternize with him, because he was treacherous. In the midst of an apparently friendly conversation, he would suddenly unveil his authority with some biting and belittling remark. He never let anyone forget he was captain.

"Robwood was afraid of him and hated him intensely. Robwood had told me privately he intended to ask for a transfer to another ship after this hop to Mars. Makki held Robwood in considerable scorn because Robwood is a timid man, and a slow thinker outside his own field of engineering. Makki made no effort to conceal that scorn.

"Taat was as contemptuous of Makki as Makki was of Robwood. Makki was ruthless with any open attempt to question his judgment, but Taat could do it with a raised eyebrow, his tone of voice or a well-chosen phrase. Makki sensed this, and alternated between treating Taat as more of an equal than either Robwood or me and 'riding' Taat harder than anyone else.

"Robwood and Taat have been aboard with us for the last five hops, but I've been with Makki since both of us graduated from the Space Academy. We were boys together, but I have never liked Makki. He always had too little respect for human dignity. He was a good space captain because he was a genius with such impersonal things as machinery and astrogation, and I have never known him to slip up on a record or let a ship get a single second off course. But mankind is better off without him."

Lefler signed off and laid the microphone down. He realized suddenly that he was perspiring and his hands were trembling. The statement had been a major emotional strain.

Unstrapping himself from the control chair, he floated down past the astrogation deck and looked in on the centerdeck. Both Taat and Robwood were strapped to their bunks, apparently asleep.

Satisfied, Lefler returned to the control room. He wanted to listen, without embarrassing interruptions, to Taat's and Robwood's statements as he transferred them from the radio recording tape to the ship's log.

The tapes rolled on the two connected machines, the log tape slowly, the radio tape at a faster clip. A loudspeaker was plugged into the radio-tape machine. Lefler kept it turned low, though the centerdeck was two decks down.

"I woke Makki at 0930 hours." It was Robwood's low voice on the tape. "He relieved me right at 1000 hours. I went down to the centerdeck and had a late lunch. Lefler strapped himself in for his sleeping period while I was eating. Taat ate lunch with me, and then we played cards for about an hour. We do that almost every day when Taat's sleeping periods are on the same schedule as mine. He changes his, because he's a psychologist and wants to watch all the crew members.

"I check the rocket engines and the fuel tanks every twenty-five days. When the Earth transit is coming up, I always do it two days ahead of time in case there are any corrections to be made in the ship's orbit. I got into a spacesuit and spent the rest of my free period outside the personnel sphere doing that. I took a break for supper, I'd say about 1600 hours, and went back to my

inspection. Taat ate with me and Lefler was asleep. Makki didn't eat with us. He did sometimes, but not often. He usually wanted to eat alone. With the Earth transit about due, I figured he'd already eaten and gone back to the control room.

"I was late for my sleeping period, but I wanted to finish my inspection. I had just gotten back through the airlock and was taking my spacesuit off when I heard Lefler call from the control room. He and Taat were both there when I got there.

"I didn't like Makki, but neither did Taat and Lefler. I suppose it'll come out, so I might as well tell about it. Makki broke up my engagement with a girl back on Earth several years ago. I wasn't going to sign on for the Mars hop because I was going to get married. Makki couldn't find an engineer to replace me, and he smooth-talked her out of it. He told me about it a long time afterward and laughed at me. I haven't ever seen her again.

"Lefler and Taat are both decent fellows and I don't think either one of them killed Makki. I think he shot himself. He ought to have!"

Robwood's final words were spoken in an outburst of concentrated bitterness. Lefler stared thoughtfully at the unwinding tapes as he waited for Taat's report to tune in. He hadn't known that about Robwood's fiancée, but it was the sort of thing Makki wouldn't hesitate to do.

"The last time I saw Makki," came Taat's calm, controlled voice from the loudspeaker, "was 1615 hours. He had just finished lunch and was going back to the control room when I came onto the centerdeck from the storage deck below. Robwood came up from below a couple of minutes later and we ate supper together.

"Robwood and I usually play a round of cards after supper when we're on the same schedule, but he was busy and I was in the middle of an experiment in the lab I have set up on the storage deck. We went down to the storage deck together. He went on below to the airlock and I started the moving picture camera again on my experiment.

"I didn't go up again until Lefler sounded the alarm. He was alone with Makki in the control room when I got there, and Makki was dead.

"I must admit it is my personal feeling that whichever of my colleagues killed Makki is a benefactor to the human race, and I hope he escapes

punishment. I did not know Makki before Robwood and I signed up together on the *Marsward IV* five voyages ago. I made the mistake of entering into a business transaction with him on our first Mars trip. He needed my capital and we became partners in purchasing a block of stock in a private dome enterprise. He accused me several times afterward of cheating him, but he handled the dividends and I think he was cheating me.

"As a psychologist, I would say that Lefler is more likely to have killed Makki coldly and deliberately, but Robwood is more likely to have killed him in the heat of an argument."

Taat's voice stopped. Lefler turned off the machines and disconnected them.

An argument. He had heard shouting in his dreams. Was that what had awakened him?

He tried to bring the dream into focus. It barely eluded him. All he could remember was that it was something about Makki.

Both Taat and Robwood were up by 0400 hours. They brought their breakfasts to the control room, along with coffee for Lefler.

It was a pleasant meal for the three of them. No one really seemed to care that one of the others was a murderer, Lefler thought. They talked and acted more like companions in crime—or like the murderer was none of them, but someone lurking somewhere else in the ship.

He wished he did not feel impelled to find out, if he could, who killed Makki. But he knew that Taat would be trying to find out, too—if Taat hadn't done it—because Taat was a psychologist and would look at it as a scientific problem. Robwood was the only one who might be temperamentally inclined to let the solution wait until they reached Mars.

When Robwood took over duty watch at 0600 hours, Lefler found Taat listening to a tape on criminal psychology on the centerdeck.

"Taat, didn't I hear you say you were working on some sort of an experiment on the storage deck while Makki was on watch yesterday?" asked Lefler.

Taat switched off the player.

"That's what I was doing," he said carefully, "but I don't remember saying anything about it."

"I listened to the reports you and Robwood made while I was recording them in the log," admitted Lefler. "I was interested in your estimate of Robwood's and my comparative abilities to commit murder."

Taat removed his spectacles, polished them and put them in his breast pocket before answering.

"I'm not surprised that you listened, Lefler—whether you're guilty or innocent," said Taat. "You probably noted that I mentioned I was recording my experiments on film. If you'll go below with me, I'd like for you to see that film."

Together, they pulled themselves down to the storage deck. Over near the main electrical switchboard, Robwood had torn out three empty spacesuit lockers and built a compact laboratory for Taat. A dozen white mice and some hamsters floated in cages attached to the wall.

For Taat's convenience, Robwood had moved the storage deck chronometer from the other side of the deck to the lab. It read 0607.

Taat unrolled a screen against one of the spacesuit lockers, attached the film roll to the projector, darkened the deck and began the showing.

The film began on Taat's face, blurred and enormously enlarged, as he switched on the camera. Taat stepped backward until he was in focus, and picked up the microphone that tied into the sound track.

"This is an experiment with white mice in a maze under conditions of zero gravity," said the Taat on the screen. Stepping aside, he waved a hand at a wire contraption on a table. "I have here a three-dimensional maze. The chronometer is visible above it, so we can check the reaction time."

Lefler noted the chronometer reading. It was 1500. In the "day" square just below its center was the figure 73.

Lefler checked the chronometer in the picture as the film ran on. There was an announced break between 1612 and 1654. Other than that, it ran continuously to 1851, when his own voice sounded faintly, calling, "Blue alert! All hands to control room. Blue alert!" At that, Taat's startled face loomed up again before the lens and the film stopped abruptly.

Throughout the approximately three hours, Taat was always in the camera's view, running his mice through the maze and explaining his methods.

"What was that forty-minute break, Taat?" asked Lefler when Taat switched the lights on once again.

"Supper," said Taat. "Robwood and I ate together, and came back down from the centerdeck together. I saw Makki leave the centerdeck when I went up, but Robwood got there a minute or two later and I don't think he saw Makki."

"You seem to have established a pretty good alibi," said Lefler slowly. "How about Robwood?"

"Lefler, for your sake, I hate to say this. The only time Robwood was above the storage deck from the time I started this film was when we had supper together. I'd have seen him if he'd passed through, and the only way he could have gotten into the control room would have been through one of the ports."

"He couldn't, without breaking it and setting off an alarm," said Lefler. "Are you trying to tell me you think I killed Makki, Taat?"

"I was here," said Taat, waving his hand at the projector. "I was between Robwood and the control room all the time. You're the only one who could have gotten there without my seeing you, Lefler, and I found you alone with him fifteen minutes after he died."

"You're sure about that fifteen minutes?"

"Within a pretty narrow range. The dilation of the pupils is an accurate gauge. I don't say you killed him, Lefler. I hope they rule it was suicide."

Silently, Lefler went back to the centerdeck, undressed and strapped himself into his bunk. He found it hard to get to sleep. Something was nagging at the back of his mind. He hoped he wouldn't dream of Makki again.

When Lefler assumed his duty watch at 1800, he asked Robwood to stay in the control room with him for a talk. Robwood strapped himself in the

astrogator's chair and waited while Lefler made the position readings. Then Lefler swung his chair around to face Robwood.

"I want to check some things with you, Robwood," he said. "I've listened to your report and Taat's and I've seen a film of Taat's that seems to give you both an alibi. After Makki relieved you and you ate lunch, was suppertime the only time you came back into the personnel sphere?"

"That's right," said Robwood. "Taat and I played cards a while after lunch, but I think you were awake then."

"How long did your supper period last?"

"Oh, half an hour. Maybe a little longer. You were asleep and snoring."

Lefler shook his head savagely.

"Robwood, I'm afraid you're going to have to take over the ship. I want you to put me in irons and turn me over for Makki's murder when we get to Marsport."

Robwood started so violently he almost broke his retaining straps. He stared at Lefler for a full thirty seconds before he found his voice.

"You're not serious!" he exclaimed. There was a pleading note to his tone. "Lefler, you didn't shoot him, did you?"

"I must have, Robwood. But not consciously. I've been able at last to remember a nightmare I had just before I found Makki's body.

"Makki and I were boys together, and he was just as mean and evil then as he was when he grew up. I was dreaming about the time Makki smashed my toy electric train and laughed about it. I tried to kill him then. I beat him with the semaphore and cut his face all up before he knocked me down and kicked me half senseless. I lived through that experience again in my dream.

"My bunk straps were loose when I woke up. I must have acted that dream out in a semi-conscious state. I must have gone up to the control room, tackled Makki and finally shot him."

"That's the silliest thing I ever heard of," retorted Robwood.

"It must be true, Robwood. Neither you nor Taat could have killed him, and Taat's got the film to prove it."

Robwood unstrapped himself and pushed himself to the companionway with some determination.

"Well, I'm not going to take over the ship and I'm not going to put you in irons," he said spiritedly. "I couldn't handle the ship on a twenty-four-hour basis for the next hundred and eighty-six days, and I'd rather think Makki killed himself."

He paused at the top of the companionway.

"Don't forget," he said. "The Earth transit ought to be at midpoint in a couple of hours."

Then he disappeared below.

Lefler took the magnetized pencil from the memorandum pad and wrote a reminder: "E.T. midpoint. Should check 28:16:54."

Lefler leaned back gloomily in the control chair. Had he killed Makki? It seemed the only way it could have happened, unless Makki had, indeed, committed suicide. And he just didn't think Makki had.

The chronometer said 1839. Exactly twenty-four hours ago, he had awakened from a nightmare and had come up to find Makki dead in this same chair. It seemed a century.

He glanced idly back at the memorandum pad. 28:15:64. He'd have to make an entry in the log in a little under two hours. How could he check accurately when the time of entry into transit was estimated?

Twenty-four plus two. Twenty-six.

He sat bolt upright, straining at his straps. He snapped down the communicator button.

"Robwood, come back up here!" he bellowed.

Unbuckling himself hastily, Lefler headed across the room toward the heat-gun rack.

Taat was playing solitaire, waiting patiently for Robwood, when Lefler and Robwood came down to the centerdeck together.

Lefler pointed a heat-gun at Taat.

"Go below and get the irons, Robwood," he said. "Taat, I'm sorry, but I'm arresting you for the murder of Makki."

Taat raised an eyebrow and continued shuffling cards.

"I don't think you want to do anything like that, Robwood," he said mildly. "Do you?"

Robwood hesitated and cast an anxious glance at him, but turned and headed for the companionway to the storage deck.

"You've convinced him, have you, Lefler?" said Taat. "I didn't believe you were guilty, but this makes me think you are."

Lefler said nothing, but held the gun steadily on Taat. Taat appeared relaxed, but Lefler sensed a tension in him.

"What makes you think I did it, Lefler?" sparred Taat. The light glinted from his spectacles as he turned his eyes from Lefler's face to watch the shuffling cards.

"Two things," said Lefler. "If I'd killed him in a half-asleep daze, I wouldn't have put gloves on him to make it look like suicide. Second, your film started at 1500—a strangely precise hour—and Makki was killed before then."

"The first point is good psychology," conceded Taat. "Since Robwood couldn't have done it, I'll admit it looks like suicide. But your second point doesn't hold water. Medical examination is accurate almost to a fine point on the time of death so soon afterward."

"Medical evidence may not lie, but the examiner can, Taat," said Lefler.

The clank of the chains resounded up the companionway. Robwood was coming back. The spring in Taat uncoiled.

With a single sweep, he hurled the deck of cards at Lefler's head and surged upward. Lefler lost his balance and fell sidewise as he dodged the improvised missile. But even as he lost his equilibrium, he pressed the trigger of the heat-gun and brought it downward in a fast chop.



The straps that held Taat to his chair were his doom. The searing beam swept across them, freeing him but at the same time blasting a six-inch swath across his stomach. Taat screamed hoarsely as the beam swung past him and burned along the floor of the centerdeck.

Lefler regained his balance and floated to Taat's side, pushing aside the cards that drifted in a swirling cloud about the room. Robwood appeared from below, the manacles in his hands.

"Your third point wins the day," gasped Taat, his hands writhing over his mangled abdomen. "I won't last long, but if you'll get me to the control room I'll radio a confession that'll clear you and Robwood completely."

"Help me get him to a bunk, Robwood," ordered Lefler, grasping Taat by the arms. "Taata, you'll have to tell us what to do for you."

"No use," groaned Taat. He managed a ghastly smile. "I unbuckled your bunk straps to throw you off course, Lefler, but I don't want you to think I was trying to blame it on you. I was trying to make it look like Makki killed himself."

"But why, Taat?"

"It wasn't just that Makki cheated me," replied Taat with some difficulty. "I'd saved several thousand dollars to build a little clinic in Mars City—something I've dreamed of all my life. That's why I let Makki talk me into investing—I needed just a little more. But the business was almost worthless. He stole most of my money. I was arguing with him about it in the control room, when he drew the gun and threatened to kill me. He was strapped down. I wrestled with him, and he was killed in the scuffle. That's it."

They maneuvered Taat into a bunk and tried to arrange the straps to avoid the gaping wound in his stomach. Taat raised his hand weakly and removed his spectacles. He blinked up at Lefler.

"I didn't think you knew enough about medicine to tell how long a man had been dead," he said.

"I don't," said Lefler. "But you set the time of Makki's death at 1830 hours. You said you could tell."

"The Earth transit started at 1612, Taat. I've known Makki all my life. If he'd been alive then, he'd have recorded it in the log. And he didn't."

"I just figured the only man who had any reason to lie deliberately about the time of Makki's death was the man who shot him."

Lefler looked at the centerdeck chronometer. It was 2025.

"Do what you can for him, then bring him up to the radio, Robwood," he said. "I've got to get up to the control room and record the midpoint of the Earth transit."

*** END OF THE PROJECT GUTENBERG EBOOK EARTH TRANSIT

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.